

12.6 Cylinders and Quadric Surfaces

1 Warm Up

Problem 1: Graph the curve $y = 9 - x^2$.

Question 2: What is an ellipse? What is the equation of an ellipse?

2 Motivation

In 2-dimension we spend many years of math classes learning how to graph curves on the xy plane. In order to understand the tools of calculus in 3 dimensions, we similarly need to be proficient at graphing and visualizing surfaces in the xyz coordinate system.

3 Definitions & Theorems

1. If we just took a screenshot of a surface on any plane parallel to our coordinate planes (xy,xz,yz) we would get a picture which looks like a curve in 2-dimensions. We call this a **trace**. A trace can be thought of as a cross section.
2. A **cylinder** is a surface which has 2 variables and thus, its trace remains unchanged as it moves through the axis of the variable missing from the surface.
3. A **Quadric Surface** is a surface with largest degree 2. The most general equation for this is,

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0$$

These will be some of the surfaces we *do* want to feel comfortable drawing in 3-dimensions.

4. An **Ellipsoid** is of the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

Its x -intercepts are at $\pm a$, its y -intercepts are at $\pm b$ and its z -intercepts are at $\pm c$. Here are ways to identify the ellipsoid.

- ALL positive.
- All variables squared.

- Has a constant.

5. A **1-sheet Hyperboloid** is of the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1.$$

It always is in the direction of the variable with the minus sign. Here are ways to identify the 1-sheet hyperboloid.

- Has one minus sign.
- All variables squared.
- Has a constant.

To start graphing set the negative variable equal to 0 and $\pm\sqrt{\text{denominator}}$ for 3 traces (which will always be circles or ellipses).

6. A **2-sheet Hyperboloid** is of the form

$$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

It always is in the direction of the variable with the plus sign. Here are ways to identify the 2-sheet hyperboloid.

- Has one plus sign.
- All variables squared.
- Has a constant.

To start graphing set both the negative variable equal to 0 to find the intercepts of the axis. Set the plus variable equal to some numbers divisible by the denominator.

7. A **Cone** is of the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0.$$

It always is in the direction of the variable with the minus sign. Here are ways to identify the 1-sheet hyperboloid.

- Has one minus sign.
- All variables squared.
- NO constant.

Our cone will hit the origin. To graph traces, set negative variable equal to numbers which are divisible by the denominator (traces will be circles and ellipses).

8. A **Paraboloid** is of the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = cz$$

To tell we have a paraboloid check for the following.

- 3 variables 2 of which are squared
- squared variables are positive

Paraboloids travel in the direction of the degree one variable and opens in the direction depending on the variable c (positive or negative). To graph set the degree one variable equal to 0 (unless shifted).

9. A **Hyperbolic Paraboloid** is of the form

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = cz$$

To tell we have a paraboloid check for the following.

- 3 variables 2 of which are squared
- one squared variables is positive

Paraboloids travel in the direction of the degree one variable. Plug in positives and negatives into the degree one variable to check whether we get hyperbolas on certain axis.

4 Examples

Problem 1. *Sketch the surface $z = 9 - x^2$.*

Problem 2. *Sketch the surface $yz = 1$.*

Problem 3. *Sketch the surface $4x^2 + 16y^2 + z^2 = 64$.*

Problem 4. *Sketch the curve $9x^2 + 9z^2 - 4y^2 = 36$.*

Problem 5. *Sketch the curve $-z^2 + 9x^2 + y^2 + 9 = 0$*

Problem 6. *Sketch the surface $y^2 = 4x^2 + 16z^2$.*

Problem 7. *Sketch the surface $z = x^2 + 4y^2 - 4$.*

Problem 8. *Sketch the surface $x = 4 - 5y^2 - 9z^2$.*

Problem 9. *Sketch the surface $x^2 - y^2 - z = 0$.*

5 Scratch Work